

1.0 KNOW SIMPLE FILE ORGANIZATION CONCEPT

1.1 Explain File Organization and Management

File organization: refers to the way data is stored in a file. File organization is very important because it determines the methods of access, efficiency, flexibility and storage devices to use.

File management: is a process by which computer user follow to keep track of different files. Unlike the CPU and memory management aspects of the operating system which aim mainly towards an optimum use of the CPU, file management aims to provide a convenient programming environment for the users of the system. In fact file management is often used by the users as a measure of how good the operating system is.

1.2 Explain the concept of file organization in computing

A file could be defined as contiguous set of words (bits, bytes or instructions). More appropriately. A file is collection of data. We can store different kinds of information in a file. For example source codes, images. Text documents and so forth. Files are of different types depending on the information stored in a file and therefore each file has its own structure which is type dependent. Everything on a computer is stored as a file. It is the job of the operating system to map this sequence of words (i.e. files) into physical devices. This gives us another alternative definition of a file system; the part of operating system that maps various files to the physical devices is called file organization and management system.

1.3 Explain the concept of record, field, character, byte and bits in relation to a file

Record: Record is the collection of organizes and related fields.

Record - a record represents a collection of attributes that describe a real-world entity. A record consists of fields, with each field describing an attribute of the entity. File - a group of related records. Files are frequently classified by the application for which they are primarily used (employee file).

Field: A field is made up of characters, or simply defined as a combination of characters. It's an attribute that can be assigned values. For example, First name, Age, Gender, Date of birth, etc. Field - a field consists of a grouping of characters. A data field represents an attribute (a characteristic or quality) of some entity (object, person, place, or event). Record - a record represents a collection of attributes that describe a real-world entity.

Character: A character is the smallest unit of information. It includes letters, digits and special characters such as; +, -, *, %, &, etc.

A character is the smallest element in a computer file and refers to letter, number or symbol that can be entered, stored and output by a computer. A character is made up of seven or eight bits depending on the character coding scheme used.

Byte is the smallest addressable unit of memory. For example, an 820 megabyte (MB) hard drive holds a nominal 820 million bytes of data. Due to massive increases in storage capacity over time, there are now eight additional units of measurement following the byte.

Bit is a binary digit, the smallest increment of data on a computer. A bit can hold only one of two values: 0 or 1, corresponding to the electrical values of off or on, respectively. Because bits are so small, you rarely work with information one bit at a time.

File: File is the collection of organized and related records.

2.0: UNDERSTAND THE CONCEPT OF FILE OPERATIONS

2.1 Different methods of file organization in computer system

Various methods have been introduced to Organize files. These particular methods have advantages and disadvantages on the basis of access or selection. Thus it is all upon the programmer to decide the best-suited file Organization method according to his requirements.

- ❖ Sequential File Organization
- ❖ Heap File Organization
- ❖ Hash File Organization
- ❖ B+ Tree File Organization
- ❖ Clustered File Organization
- ❖ ISAM (Indexed Sequential Access Method)

1) SEQUENTIAL FILE ORGANIZATION: - The easiest method for file Organization is the Sequential method. In this method, the file is stored one after another in a sequential manner. There are two ways to implement this method:

Pile File Method: This method is quite simple, in which we store the records in a sequence i.e. one after the other in the order in which they are inserted into the tables.

Insertion of the new record: Let the R1, R3, and so on up to R5 and R4 be four records in the sequence. Here, records are nothing but a row in any table. Suppose a new record R2 has to be inserted in the sequence, and then it is simply placed at the end of the file.

2. Sorted File Method: In this method, as the name itself suggests whenever a new record has to be inserted, it is always inserted in a sorted (ascending or descending) manner. The sorting of records may be based on any primary key or any other key.

Insertion of the new record: Let us assume that there is a preexisting sorted sequence of four records R1, R3, and so on up to R7 and R8. Suppose a new record R2 has to be inserted in the sequence, then it will be inserted at the end of the file and then it will sort the sequence.

Advantages of Sequential File Organization

- ❖ Fast and efficient method for huge amounts of data.
- ❖ Simple design.
- ❖ Files can be easily stored in magnetic tapes i.e. cheaper storage mechanism.

Disadvantages of Sequential File Organization

- ❖ Time wastage as we cannot jump on a particular record that is required, but we have to move in a sequential manner which takes our time.
- ❖ The sorted file method is inefficient as it takes time and space for sorting records.

2) HEAP FILE ORGANIZATION: - works with data blocks. In this method, records are inserted at the end of the file, into the data blocks. No Sorting or Ordering is required in this method. If a data block is full, the new record is stored in some other block, here the other data block need not be the very next data block, but it can be any block in the memory. It is the responsibility of DBMS to store and manage the new records.

Insertion of the new record: Suppose we have four records in the heap R1, R5, R6, R4, and R3, and suppose a new record R2 has to be inserted in the heap then, since the last data block

i.e data block 3 is full it will be inserted in any of the data blocks selected by the DBMS, let's say data block 1.

If we want to search, delete or update data in the heap file Organization we will traverse the data from the beginning of the file till we get the requested record. Thus if the database is very huge, searching, deleting, or updating the record will take a lot of time.

Advantages of Heap File Organization

- ❖ Fetching and retrieving records is faster than sequential records but only in the case of small databases.
- ❖ When there is a huge number of data that needs to be loaded into the database at a time, then this method of file Organization is best suited.

Disadvantages of Heap File Organization

- ❖ The problem of unused memory blocks.
- ❖ Inefficient for larger databases.

3) HASHING FILE ORGANIZATION: - is an efficient technique to directly search the location of desired data on the disk without using index structure. Data is stored at the data blocks whose address is generated by using hash function. The memory location where these records are stored is called as data block or data bucket.

In a database management system, when we want to retrieve a particular data, it becomes very inefficient to search all the index values and reach the desired data. In this situation, Hashing technique comes into picture.

Hash data structures have many applications in computer science, including implementing symbol tables, caches, and databases. They are especially useful in situations where fast retrieval of elements is important, and where the number of elements to be stored may be large.

Collision Resolution: Collision resolution in hash can be done by two methods:

- ❖ Open addressing
- ❖ Closed addressing.

Open Addressing: Open addressing collision resolution technique involves generating a location for storing or searching the data called **probe**. It can be done in the following ways:

- **Linear Probing:** If there is a collision at **i** then we use the hash function – $H(k, i) = [H'(k) + i] \% m$

where, **i** is the index, **m** is the size of hash table $H(k, i)$ and $H'(k)$ are hash functions.

- **Quadratic Probing:** If there is a collision at **i** then we use the hash function – $H(k, i) = [H'(k) + c_1 * i + c_2 * i^2] \% m$

where, **i** is the index, **m** is the size of hash table $H(k, i)$ and $H'(k)$ are hash functions, c_1 and c_2 are constants.

- **Double Hashing:** If there is a collision at **i** then we use the hash function – $H(k, i) = [H_1(k, i) + i * H_2(k)] \% m$

where, **i** is the index, **m** is the size of hash table $H(k, i)$, $H_1(k) = k \% m$ and $H_2(k) = k \% m'$ are hash functions.

Closed Addressing: Closed addressing collision resolution technique involves chaining. Chaining in the hashing involves both array and linked list. In this method, we generate a probe with the help of the hash function and link the keys to the respective index one after the other in the same index. Hence, resolving the collision.

Advantage of Hash:

- ❖ Hash provides better synchronization than other data structures.

- ❖ Hash tables are more efficient than search trees or other data structures.
- ❖ Hash provides constant time for searching, insertion and deletion operations on average.
- ❖ Hash tables are space-efficient.
- ❖ Most Hash table implementation can automatically resize itself.
- ❖ Hash tables are easy to use.
- ❖ Hash tables offer a high-speed data retrieval and manipulation.

Disadvantages of Hash:

- ❖ Hash is inefficient when there are many collisions.
- ❖ Hash collisions are practically not be avoided for large set of possible keys.
- ❖ Hash does not allow null values.
- ❖ Hash tables have a limited capacity and will eventually fill up.
- ❖ Hash tables can be complex to implement.
- ❖ Hash tables do not maintain the order of elements, which makes it difficult to retrieve elements in a specific order.

4) B+ TREE: - as the name suggests, It uses a tree-like structure to store records in a File. It uses the concept of Key indexing where the primary key is used to sort the records. For each primary key, an index value is generated and mapped with the record. An index of a record is the address of the record in the file.

B+ Tree is very much similar to a binary search tree, with the only difference being that instead of just two children, it can have more than two. All the information is stored in a leaf node and the intermediate nodes act as a pointer to the leaf nodes. The information in leaf nodes always remains a sorted sequential linked list.

Advantages of B+ Tree File Organization

- ❖ Tree traversal is easier and faster.
- ❖ Searching becomes easy as all records are stored only in leaf nodes and are sorted in sequentially linked lists.
- ❖ There is no restriction on B+ tree size. It may grow/shrink as the size of the data increases/decreases.

Disadvantages of B+ Tree File Organization

- ❖ Inefficient for static tables.

5) CLUSTER FILE ORGANIZATION: - In **Cluster file organization**, two or more related tables/records are stored within the same file known as clusters. These files will have two or more tables in the same data block and the key attributes which are used to map these tables together are stored only once.

Thus it lowers the cost of searching and retrieving various records in different files as they are now combined and kept in a single cluster. For example, we have two tables or relation Employee and Department. These tables are related to each other. Therefore this table is allowed to combine using a join operation and can be seen in a cluster file.

Types of Cluster File Organization

There are two ways to implement this method.

Indexed Clusters: In Indexed clustering, the records are grouped based on the cluster key and stored together. The above-mentioned example of the Employee and Department relationship is an example of an Indexed Cluster where the records are based on the Department ID.

Hash Clusters: This is very much similar to an indexed cluster with the only difference that instead of storing the records based on cluster key, we generate a hash key value and store the records with the same hash key value.

Advantages of Cluster File Organization

- ❖ It is basically used when multiple tables have to be joined with the same joining condition.

- ❖ It gives the best output when the cardinality is 1:n.

Disadvantages of Cluster File Organization

- ❖ It gives a low performance in the case of a large database.
- ❖ In the case of a 1:1 cardinality, it becomes ineffective.

2.2 The different file operations; storage, retrieval, add delete, update and maintenance.

- ❖ **File storage:** - also called file-level or file-based storage, is a type of IT storage for storing data in a hierarchical structure. Data is saved in files inside folders, nested within other folders, analogous to an office file cabinet.
- ❖ **Retrieval:** - Searching for, locating, and returning data. For example, a user may retrieve a document on a computer to be viewed or modified.
- ❖ **Update:** - An update file is an input file whose records can be read and updated. Updating alters the data in one or more fields of any record contained in the file and writes that record back to the same file from which it was read.
- ❖ **Maintenance:** - The routine changes, updates, copying, moving, or deleting of files on a computer. Usually, **file maintenance** is performed on computers or servers that are serving a vast amount of files.

2.3 Activity ratio and hit rate.

- ❖ **Activity Ratios:** - Activity ratios are used to determine the efficiency of the organization in utilizing its assets for generating cash and revenue. It is used to check the level of investment made on an asset and the revenue that it is generating.
- ❖ **Hit rate:** - is a metric or measure of business performance traditionally associated with sales. It is defined as the number of sales of a product divided by the number of customers who go online, planned call, or visit a company to find out about the product

2.4 Different types of files: Master file, Transaction file, Reference file, etc.

There are numerous types of files used for storing data needed for processing, reference or back up. The main common types of processing files include

- ❖ Master files,
- ❖ Transaction,
- ❖ Reference,
- ❖ Backup, report and
- ❖ Sort file.

Master file: - A master file is the main that contains relatively permanent records about particular items or entries. For example a customer file will contain details of a customer such as customer ID, name and contact address.

Transaction (movement) file

A transaction file: - is used to hold data during transaction processing. The file is later used to update the master file and audit daily, weekly or monthly transactions. For example in a busy supermarket, daily sales are recorded on a transaction file and later used to update the stock file. The file is also used by the management to check on the daily or periodic transactions.

Reference file: - A reference file is mainly used for reference or look-up purposes. Look-up information is that information that is stored in a separate file but is required during processing. For example, in a point of sale terminal, the item code entered either manually or using a barcode reader looks up the item description and price from a reference file stored on a storage device.

Backup file: - A backup files is used to hold copies (backups) of data or information from the computers fixed storage (hard disk). Since a file held on the hard disk may be corrupted, lost or changed accidentally, it is necessary to keep copies of the recently updated files. In case of the hard disk failure, a backup file can be used to reconstruct the original file.

Report file: - Used to store relatively permanent records extracted from the master file or generated after processing. For example you may obtain a stock levels report generated from an inventory system while a copy of the report will be stored in the report file.

Sort file: - It stores data which is arranged in a particular order. Used mainly where data is to be processed sequentially. In sequential processing, data or records are first sorted and held on a magnetic tape before updating the master file.

2.5 The concept of master file, transaction file and reference file.

- ❖ **Master file:** there are files of a fairly permanent nature, e.g. customer ledger, payroll, inventory, and so on. A feature to know is the regular updating of these files to show a current position. For example, customer's order will be processed, increasing the "balance owing" figure on a customer ledger record. It is seen therefore that master records will contain both data of a static nature, e.g. a customer name, address, and data that, by its nature will change each time a transaction occurs, e.g. the "balance" figure already mentioned.
- ❖ **Transaction file:** This is also known as movement file. This is made up of various transactions created from the source documents. In a sales ledger application the file will contain all the orders received at a particular time. This file will be used to update the master file. As soon as it had been used for this purpose it is no longer required. It will therefore have a very short life, because it will be replace by a file containing the next batch of orders.
- ❖ **Reference files:** A file with a reasonable amount of permanency. Examples of data used for reference purposes are price lists, tables of rates of pay, names and addresses.

Criteria for classifying computer files are:

- ❖ By nature of content: it refers to the nature of file content.
- ❖ By organization method: it refers to the way files are arranged e.g. Serial, sequential, random and so on.

- ❖ By storage medium: it refers to storage devices in which a file's could only be stored such as magnetic or optical disk and magnetic tape and so on.

3.0: UNDERSTAND THE BASIC STORAGE DEVICES AND MEDIA.

3.1 Types of storage devices and media

Types of Computer Storage Devices

Now we will discuss different types of storage devices available in the market. These storage devices have their own specification and use. Some of the commonly used storage devices are:

1. Primary Storage Devices
2. Magnetic Storage Devices
3. Flash memory Devices
4. Optical Storage Devices
5. Cloud and Virtual Storage

1) Primary Storage Devices

- **RAM:** It stands for Random Access Memory. It is used to store information that is used immediately or we can say that it is a temporary memory. Computers bring the software installed on a hard disk to RAM to process it and to be used by the user. Once, the computer is turned off, the data is deleted. With the help of RAM, computers can perform multiple tasks like loading applications, browsing the web, editing a spreadsheet, experiencing the newest game, etc. It allows you to modify quickly among these tasks, remembering where you're in one task once you switch to a different task. It is also used to load and run applications, like your spreadsheet program, answers commands, like all edits you made within the spreadsheet, or toggle between multiple programs, like once you left the spreadsheet to see the email. Memory is nearly always actively employed by your computer. It ranges from 1GB – 32GB/64GB depending upon the specifications. There are different types of RAM, and although they all serve the same purpose, the most common ones are :

- **SRAM:** It stands for Static Random Access Memory. It consists of circuits that retain stored information as long as the power supply is on. It is also known as volatile memory. It is used to build Cache memory. The access time of SRAM is lower and it is much faster as compared to DRAM but in terms of cost, it is costly as compared to DRAM.
- **DRAM:** It stands for Dynamic Random Access Memory. It is used to store binary bits in the form of electrical charges that are applied to capacitors. The access time of DRAM is slower as compared to SRAM but it is cheaper than SRAM and has a high packaging density.
- **SDRAM:** It stands for Synchronous Dynamic Random Access Memory. It is faster than DRAM. It is widely used in computers and others. After SDRAM was introduced, the upgraded version of double data rate RAM, i.e., DDR1,

DDR2, DDR3, and DDR4 was entered into the market and widely used in home/office desktops and laptops.

- **ROM:** It stands for Read-Only Memory. The data written or stored in these devices are non-volatile, i.e, once the data is stored in the memory cannot be modified or deleted. The memory from which will only read but cannot write it. This type of memory is non-volatile. The information is stored permanently during manufacture only once. ROM stores instructions that are used to start a computer. This operation is referred to as bootstrap. It is also used in other electronic items like washers and microwaves. ROM chips can only store a few megabytes (MB) of data, which ranges between 4 and 8 MB per ROM chip. There are two types of ROM:

- **PROM:** PROM is Programmable Read-Only Memory. These are ROMs that can be programmed. A special PROM programmer is employed to enter the program on the PROM. Once the chip has been programmed, information on the PROM can't be altered. PROM is non-volatile, that is data is not lost when power is switched off.
- **EPROM:** Another sort of memory is the Erasable Programmable Read-Only Memory. It is possible to erase the info which has been previously stored on an EPROM and write new data onto the chip.
- **EEPROM:** EEPROM is Electrically erasable programmable read-only memory. Here, data can be erased without using ultraviolet light, with the use of just applying the electric field.

2) Magnetic Storage Devices

- **Floppy Disk:** Floppy Disk is also known as a floppy diskette. It is generally used on a personal computer to store data externally. A Floppy disk is made up of a plastic cartridge and secured with a protective case. Nowadays floppy disk is replaced by new and effective storage devices like USB, etc.
 - **Hard Disk:** Hard Disk is a storage device (HDD) that stores and retrieves data using magnetic storage. It is a non-volatile storage device that can be modified or deleted a number of times without any problem. Most computers and laptops have HDDs as their secondary storage device. It is actually a set of stacked disks, just like phonograph records. In every hard disk, the data is recorded electromagnetically in concentric circles or we can say track present on the hard disk, and with the help of a head just like a phonograph arm(but fixed in a position) to read the information present on the track. The read-write speed of HDDs is not so fast but decent. It ranges from a few GBs to a few and more TB.
 - **Magnetic Card:** It is a card in which data is stored by modifying or rearranging the magnetism of tiny iron-based magnetic particles present on the band of the card. It is also known as a swipe card. It is used like a passcode(to enter the house or hotel room), credit card, identity card, etc.
 - **Tape Cassette:** It is also known as a music cassette. It is a rectangular flat container in which the data is stored in an analog magnetic tape. It is generally used to store audio recordings.
 - **SuperDisk:** It is also called LS-240 and LS-120. It is introduced by Imation Corporation and it is popular with OEM computers. It can store data up to 240 MB.
- 3) **Flash Memory Devices:** - It is a cheaper and more portable storage device. It is the most commonly used device to store data because is more reliable and efficient as compared to other storage devices. Some of the commonly used flash memory devices are:

- **Pen Drive:** It is also known as a USB flash drive that includes flash memory with an integrated USB interface. We can directly connect these devices to our computers and laptops and read/write data into them in a much faster and more efficient way. These devices are very portable. It ranges from 1GB to 256GB generally.
- **SSD:** It stands for Solid State Drive, a mass storage device like HDD. It is more durable because it does not contain optical disks inside like hard disks. It needs less power as compared to hard disks, is lightweight, and has 10x faster read and writes speed as compared to hard disks. But, these are costly as well. While SSDs serve an equivalent function as hard drives, their internal components are much different. Unlike hard drives, SSDs don't have any moving parts and thus they're called solid-state drives. Instead of storing data on magnetic platters, SSDs store data using non-volatile storage. Since SSDs haven't any moving parts, they do not need to "spin up". It ranges from 150GB to a few more TB.
- **SD Card:** It is known as a Secure Digital Card. It is generally used with electronic devices like phones, digital cameras, etc. to store larger data. It is portable and the size of the SD card is also small so that it can easily fit into electronic devices. It is available in different sizes like 2GB, 4GB, 8GB, etc.
- **Memory Card:** It is generally used in digital cameras. Printers, game consoles, etc. It is also used to store large amounts of data and is available in different sizes. To run a memory card on a computer you require a separate memory card reader.
- **Multimedia Card:** It is also known as MMC. It is an integrated circuit that is generally used in-car radios, digital cameras, etc. It is an external device to store data/information.

4) Optical Storage Devices

Optical Storage Devices is also secondary storage device. It is a removable storage device. Following are some optical storage devices:

- **CD:** It is known as Compact Disc. It contains tracks and sectors on its surface to store data. It is made up of polycarbonate plastic and is circular in shape. CD can store data up to 700MB. It is of two types:
 - **CD-R:** It stands for Compact Disc read-only. In this type of CD, once the data is written can not be erased. It is read-only.
 - **CD-RW:** It stands for Compact Disc Read Write. In this type of CD, you can easily write or erase data multiple times.
- **DVD:** It is known as Digital Versatile Disc. DVDs are circular flat optical discs used to store data. It comes in two different sizes one is 4.7GB single-layer discs and another one is 8.5GB double-layer discs. DVDs look like CDs but the storage capacity of DVDs is more than as compared to CDs. It is of two types:
 - **DVD-R:** It stands for Digital Versatile Disc read-only. In this type of DVD, once the data is written can not be erased. It is read-only. It is generally used to write movies, etc.
 - **DVD-RW:** It stands for Digital Versatile Disc Read Write. In this type of DVD, you can easily write or erase data multiple times.
- **Blu-ray Disc:** It is just like CD and DVD but the storage capacity of blu ray is up to 25GB. To run a Blu-ray disc you need a separate Blu-ray reader. This Blu-ray technology is used to read a disc from a blue-violet laser due to which the information is stored in greater density with a longer wavelength.

5) Cloud and Virtual Storage: Nowadays, secondary memory has been upgraded to virtual or cloud storage devices. We can store our files and other stuff in the cloud and the data is stored for as long as we pay for the cloud storage. There are many companies that provide cloud services largely Google, Amazon, Microsoft, etc. We can pay the rent for the amount of space we need and we get multiple benefits out of it. Though it is actually being stored in a physical device located in the data centers of the service provider, the user doesn't interact with the physical device and its maintenance. For example, Amazon Web Services offers AWS S3 as a type of storage where users can store data virtually instead of being stored in physical hard drive devices. These sorts of innovations represent the frontier of where storage media goes.

3.2 The characteristics of magnetic storage media, tape, disk, cartridge, bubble, hard disk, CDROM, DVD, floppy disks, zip disk, tape streamer, flash memory, optical disk.

- ❖ Data stored in the Memory can be changed or replaced in case of a requirement, because of the mobility of the storage devices.
- ❖ Storage Devices validate that saved data can be replaced or deleted as per the requirements because the storage devices are easily readable, writable, and rewritable.
- ❖ Storage Devices are easy and convenient to access because they do not require much skill set to handle these resources.
- ❖ The storage capacity of these devices is an extra advantage to the system.
- ❖ Storage Devices have better performance and data can be easily transferred from one device to another.

4.0 UNDERSTAND DIFFERENT FILE ACCESS METHODS AND THE BUFFERING TECHNIQUES.

4.1 Different file access types: - random access, direct access and index sequential storage methods.

1) **Sequential Access:** - It is the simplest access method. Information in the file is processed in order, one record after the other. This mode of access is by far the most common; for example, editor and compiler usually access the file in this fashion. Read and write make up the bulk of the operation on a file. A read operation -*read next*- read the next position of the file and automatically advances a file pointer, which keeps track I/O location. Similarly, for the -write *next*- append to the end of the file and advance to the newly written material.

Key points:

- ❖ Data is accessed one record right after another record in an order.
- ❖ When we use read command, it move ahead pointer by one
- ❖ When we use write command, it will allocate memory and move the pointer to the end of the file

❖ Such a method is reasonable for tape.

Advantages of Sequential Access Method:

- It is simple to implement this file access mechanism.
- It uses lexicographic order to quickly access the next entry.
- It is suitable for applications that require access to all records in a file, in a specific order.
- It is less prone to data corruption as the data is written sequentially and not randomly.
- It is a more efficient method for reading large files, as it only reads the required data and does not waste time reading unnecessary data.
- It is a reliable method for backup and restores operations, as the data is stored sequentially and can be easily restored if required.

Disadvantages of Sequential Access Method:

- If the file record that needs to be accessed next is not present next to the current record, this type of file access method is slow.
- Moving a sizable chunk of the file may be necessary to insert a new record.
- It does not allow for quick access to specific records in the file. The entire file must be searched sequentially to find a specific record, which can be time-consuming.

- It is not well-suited for applications that require frequent updates or modifications to the file. Updating or inserting a record in the middle of a large file can be a slow and cumbersome process.
- Sequential access can also result in wasted storage space if records are of varying lengths. The space between records cannot be used by other records, which can result in inefficient use of storage.
- 2) Direct Access:** - Another method is *direct access method* also known as *relative access method*. A fixed-length logical record that allows the program to read and write record rapidly, in no particular order. The direct access is based on the disk model of a file since disk allows random access to any file block. For direct access, the file is viewed as a numbered sequence of block or record. Thus, we may read block 14 then block 59, and then we can write block 17. There is no restriction on the order of reading and writing for a direct access file. A block number provided by the user to the operating system is normally a *relative block number*; the first relative block of the file is 0 and then 1 and so on.

Advantages of Direct Access Method:

- The files can be immediately accessed decreasing the average access time.
- In the direct access method, in order to access a block, there is no need of traversing all the blocks present before it.

3) Index sequential method: - It is the other method of accessing a file that is built on the top of the sequential access method. These methods construct an index for the file. The index, like an index in the back of a book, contains the pointer to the various blocks. To find a record in the file, we first search the index, and then by the help of pointer we access the file directly.

Key points:

- ❖ It is built on top of Sequential access.
- ❖ It control the pointer by using index

4) Relative Record Access: - Relative record access is a file access method used in operating systems where records are accessed relative to the current position of the file pointer. In this method, records are located based on their position relative to the current record, rather than by a specific address or key value.

Key Points of Relative Record Access:

- ❖ Relative record access is a random access method that allows records to be accessed based on their position relative to the current record.
- ❖ This method is efficient for accessing individual records but may not be suitable for files that require frequent updates or random access to specific records.
- ❖ Relative record access requires fixed-length records and may not be flexible enough for some applications.
- ❖ This method is useful for processing records in a specific order or for files that are accessed sequentially.

Advantages of Relative Record Access:

- Random Access: Relative record access allows random access to records in a file. The system can access any record at a specific offset from the current position of the file pointer.
- Efficient Retrieval: Since the system only needs to read the current record and any records that need to be skipped, relative record access is more efficient than sequential access for accessing individual records.

- Useful for Sequential Processing: Relative record access is useful for processing records in a specific order. For example, if the records are sorted in a specific order, the system can access the next or previous record relative to the current position of the file pointer.

Disadvantages of Relative Record Access:

- Fixed Record Length: Relative record access requires fixed-length records. If the records are of varying length, it may be necessary to use padding to ensure that each record is the same length.
- Limited Flexibility: Relative record access is not very flexible. It is difficult to insert or delete records in the middle of a file without disrupting the relative positions of other records.
- Limited Application: Relative record access is best suited for files that are accessed sequentially or with some regularity, but it may not be appropriate for files that are frequently updated or require random access to specific records.

5) Content Addressable Access: - Content-addressable access (CAA) is a file access method used in operating systems that allows records or blocks to be accessed based on their content rather than their address. In this method, a hash function is used to calculate a unique key for each record or block, and the system can access any record or block by specifying its key for **Keys in Content-Addressable Access:**

- ❖ Unique: Each record or block has a unique key that is generated using a hash function.
- ❖ Calculated based on content: The key is calculated based on the content of the record or block, rather than its location or address.

Advantages of Content-Addressable Access:

- Efficient Search: CAA is ideal for searching large databases or file systems because it allows for efficient searching based on the content of the records or blocks.
- Flexibility: CAA is more flexible than other access methods because it allows for easy insertion and deletion of records or blocks.
- Data Integrity: CAA ensures data integrity because each record or block has a unique key that is generated based on its content.

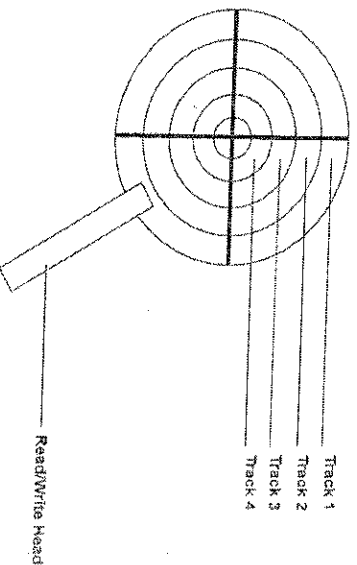
Disadvantages of Content-Addressable Access:

- Overhead: CAA requires additional overhead because the hash function must be calculated for each record or block.
- Collision: There is a possibility of collision where two records or blocks can have the same key. This can be minimized by using a good hash function, but it cannot be completely eliminated.
- Limited Key Space: The key space is limited by the size of the hash function used, which can lead to collisions and other issues.

4.2 Seek time and rotational delay

- ❖ **Seek Time:** - A disk is divided into many circular tracks. Seek Time is defined as the time required by the read/write head to move from one track to another.

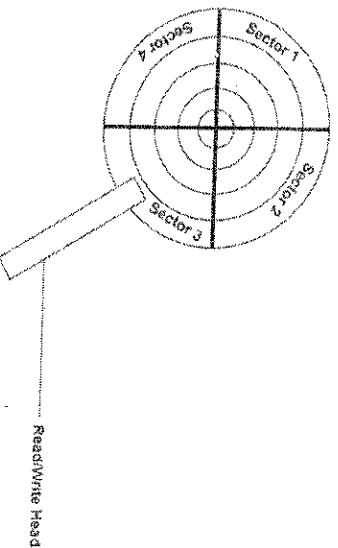
Example: Consider the following diagram, the read/write head is currently on track 1.



Now, on the next read/write request, we may want to read data from Track 4, in this case, our read/write head will move to track 4. The time it will take to reach track 4 is the **seek time**.

❖ **Rotational Latency:** -The disk is divided into many circular tracks, and these tracks are further divided into blocks known as sectors. The time required by the read/write head to rotate to the requested sector from the current position is called Rotational Latency.

Example: Consider the following diagram; we have divided each track into 4 sectors. The systems get a request to read a sector from track 1, thus the read/write head will move to track 1 and this time will be seek time. The read/write head is currently in sector 3.



But the data may not be in sector 3. The data block may be present in sector 1. The time required by read/write head to move from sector 3 to sector 1 is the **rotational latency**.

4.3 The concept of a buffer and its functions

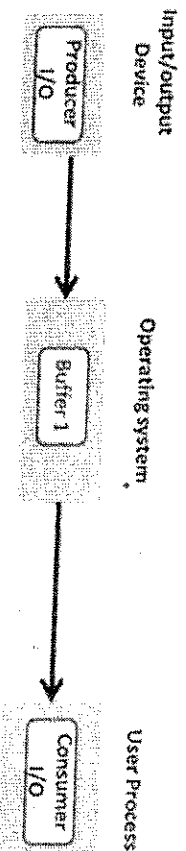
A buffer: - is a storage location that is used to store data temporarily as it transits from one device or process to the next. While buffering is the process of using buffers to increase the performance and efficiency of the system using buffers. In an operating system, buffers are used during input and output operations.

Functions of buffer in computer OS

- Buffers are used to achieve a number of objectives depending on the system.
- Smoothing the input and output operations. Buffering is used to smoothen operation on devices that have different operating speeds and data block sizes. It hides the difference between the 2 devices for smooth operation.

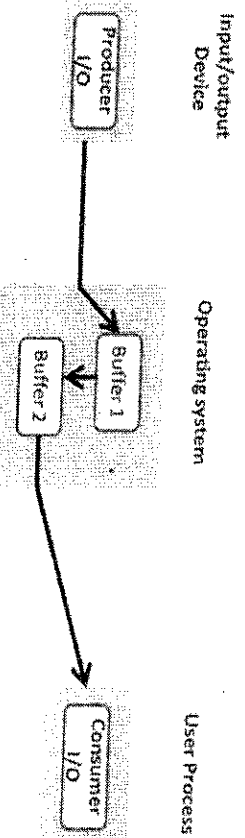
- Improve system performance. When different devices are synchronized they improve the overall system performance.
- Reduce system overhead during data transfer. This is achieved by requesting data only once then it is stored on the buffer and used continuously instead of making requests every time data is required.

SINGLE BUFFERING IN OPERATING SYSTEM



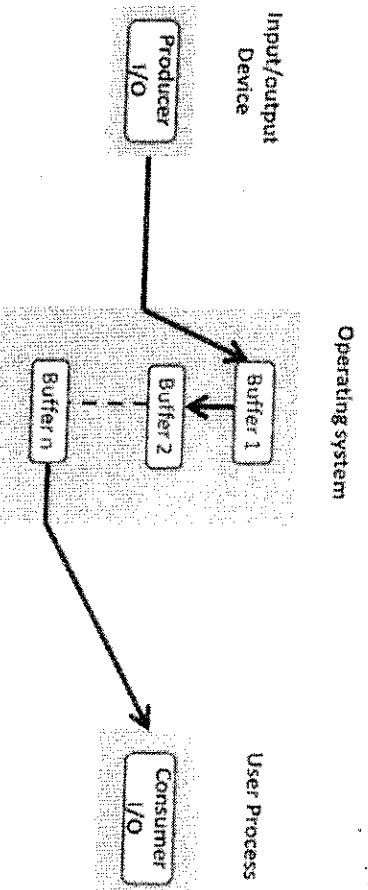
This is the simplest buffer implementation and is used in a simple system. It is implemented where the operating system allocates only one system buffer to be used. The data is moved from the source then moved to the single buffer location and transferred to the destination user space.

DOUBLE BUFFERING



This is an improvement of the single buffer where instead of OS allocating only 1 buffer they are 2. Now the data is moved from the source to the first buffer then it is moved to the second and finally to the destination device. On one side, the operating system can be filling one buffer with data, while process data is removing from the other.

CIRCULAR BUFFERING IN OS



To improve the system even further we can implement more than 2 temporally locations creating a circular buffer. This buffering is mostly used in systems that require fast operation and moving large blocks of data.

Advantages of buffers in the computer system

- Reduce the number of input/output operations required to access data in the system.
- Used to smoothen input and output operation among different devices.
- Improves the system by reducing the number of system calls made when caring out an operation.

- They can act as a caching system for the system.
- Buffering can be used for error handling by storing data with the error until it is resolved.

Disadvantages of buffers in an operating system

- The use of buffers is expensive and complex to manage. It requires high overhead to manage data stored on the buffer.
- A system with a small amount of main memory, buffers reduced the available memory for use by other processes.
- Buffer overflow. This is when more data is stored on the buffer that it can hold. This leads to the corruption of data and insecurities.

4.4 The calculation of buffer requirement of a file.

Calculating the Number of Buffers Required Based on Full-Size Frames

Use the following procedure to calculate the number of buffers required for a long-distance connection:

- Determine the desired distance in kilometers of the switch-to-switch connection.
- Determine the speed that you will use for the long-distance connection.
- Use one of the following formulas to calculate the reserved buffers for distance:
 - If QoS is enabled:

$$(\text{Reserved_Buffer_for_Distance_Y}) = (\times * \text{LinkSpeed} / 2) + 6 + 14$$

- If QoS is not enabled:

$$(\text{Reserved_Buffer_for_Distance_Y}) = (\times * \text{LinkSpeed} / 2) + 6$$

The formulas use the following parameters:

$\times$ = The distance determined in Step 1 (in km).

LinkSpeed = The speed of the link determined in Step 2.

6 = The number of buffer credits reserved for fabric services, multicast, and broadcast traffic. This number is static.

14 = The number of buffer credits reserved for QoS. This number is static.

Using 50 km as the desired distance of the switch-to-switch connection and 2G as the speed of the long-distance connection, insert the numbers into the appropriate formula. The formula should read as follows:

$$(50 \text{ km} * 2G / 2) + 6 = 56 \text{ buffers, which is the number of buffers reserved for distance.}$$

The following examples use different speeds, all based on a distance of 50 km. The distances and speeds are variables that can change depending on how your network is set up.

- If you have a distance of 50 km at 1G, $(50 \text{ km} * 1G / 2) + 6 = 31$ buffers.
- If you have a distance of 50 km at 2G, $(50 \text{ km} * 2G / 2) + 6 = 56$ buffers.
- If you have a distance of 50 km at 4G, $(50 \text{ km} * 4G / 2) + 6 = 106$ buffers.
- If you have a distance of 50 km at 8G, $(50 \text{ km} * 8G / 2) + 6 = 206$ buffers.
- If you have a distance of 50 km at 10G, $(50 \text{ km} * 10G / 2) + 6 = 256$ buffers.
- If you have a distance of 50 km at 16G, $(50 \text{ km} * 16G / 2) + 6 = 406$ buffers.
- If you have a distance of 50 km at 32G, $(50 \text{ km} * 32G / 2) + 6 = 806$ buffers.

Example

Consider the

, which has a single 24-port port group and a total of 676 buffer credits for that port group. The formulas use the following parameters:

24 = The number of user ports in a port group

8 = The number of reserved credits for each user port

676 = The number of buffer credits available in the port group

The maximum remaining number of buffer credits for the port group, after each port reserves its 8 buffer credits, is obtained from the following formula:

$$676 - (24 * 8) = 484 \text{ unreserved buffer credits}$$

492 buffers to a single port $(484 + 8)$ [8 for the reserved buffers already allocated to that user port]), you can calculate the maximum single-port extended distance supported:

$$\text{Maximum Distance } \times (\text{in km}) = (\text{BufferCredits} + 6) * 2 / \text{LinkSpeed}$$

$$498 \text{ km} = (492 + 6 \text{ buffers for Fabric Services}) * 2 / 2G$$

If you have a distance of 50 km at 8G, then $484 / (206 - 8) = 2$ ports.

The following values are used in the example:

- 484 – The total number of unreserved buffer credits
- 206 – Buffer credits needed for 50 km at 8G
- 8 – The number of reserved buffer credits already allocated to that port

The resulting number is rounded down to the next whole number because fractions of a port are not allowed.

If you have a distance of 50 km at 1G, then $484 / (31 - 8) = 21$ ports.

5.0 UNDERSTAND FILE ORGANIZATIONAL STRUCTURE AND PROCESSING.

5.1 File structure and organization

FILE STRUCTURE: - is the way files are organized on a computer. It's important because it helps you find the files you need. Adopting a file structure makes getting to the files you need as easy as possible. Specifically, it's to minimize the number of trips to the disk to get the data you're looking for.

FILE ORGANIZATION: - refers to the way in which data is stored in a file and, consequently, the method(s) by which it can be accessed. This COBOL system supports three file organizations: sequential, relative and indexed.

5.2 File processing technique

1) **Commercial Data Processing:** - means a method of applying standard relational databases, and it includes the usage of batch processing. It involves providing huge data as input into the system and creating a large volume of output but using fewer computational operations. It basically combines commerce and computers for making it useful for a business. The data that is processed through this system is usually standardized and therefore has a much lower chance of errors.

Many manual works are automated through the use of computers to make it easy and error-proof. Computers are used in business to take raw data and process it into a form of information that is useful to the business. Accounting programs are prototypical examples of data processing applications. An Information System (IS) is the field that studies such as organizational computer systems.

2) **Scientific Data Processing:** - Unlike commercial data processing, Scientific data processing involves a large use of computational operations but lower volumes of inputs as well as outputs. The computational operations include arithmetical and comparison operations. In this type of processing, any chances of errors are not acceptable as it would lead to wrongful decision-

making. Hence the process of validating, sorting, and standardizing the data is done very carefully, and a wide variety of scientific methods are used to ensure no wrong relationships and conclusions are reached.

This takes a longer time than in commercial data processing. The common examples of scientific data processing include processing, managing, and distributing science data products and facilitating scientific analysis of algorithms, calibration data, and data products as well as maintaining all software, calibration data, under strict configuration control.

3) Batch Processing: - means a type of Data Processing in which a number of cases are processed simultaneously. The data is collected and processed in batches, and it is mostly used when the data is homogenous and in large quantities. Batch Processing can be defined as the concurrent, simultaneous, or sequential execution of an activity. Simultaneous Batch processing occurs when they are executed by the same resource for all the cases at the same time. Sequential Batch processing occurs when they are executed by the same resource for different cases either immediately or immediately after one another.

Concurrent Batch processing means when they are executed by the same resources but partially overlapping in time. It is used mostly in financial applications or at places where additional levels of security are required. In this processing, the computational time is relatively less because applying a function to the whole data altogether extracts the output. It is able to complete work with a very less amount of human intervention.

4) Online Processing: - In the parlance of today's database systems, "online" signifies "interactive", within the bounds of patience." Online processing is the opposite of "batch" processing. Online processing can be built out of a number of relatively more simple operators, much as traditional query processing engines are built. Online Processing Analytical operations typically involve major fractions of large databases. It should therefore be surprising that today's Online analytical systems provide interactive performance. The secret to their success is precomputation.

In most Online Analytical Processing systems, the answer to each point and click is computed long before the user even starts the application. In fact, many Online processing systems do that computation relatively inefficiently, but since the processing is done in advance, the end-user does not see the performance problem. This type of processing is used when data is to be processed continuously, and it is fed into the system automatically.

5) Real-Time Processing: - The current data management system typically limits the capacity of processing data on an and when basis because this system is always based on periodic updates of batches due to which there is a time lag of many hours in happening of an event and recording or updating it. This caused a need for a system that would be able to record, update and process the data on as and when basis, i.e. in real-time which would help in reducing the time lag between occurrence and processing to almost nil. Huge chunks of data are being poured into systems off organizations, hence storing and processing it in a real-time environment would change the scenario.

Most organizations want to have real-time insights into the data so as to understand the environment within or outside their organization fully. This is where the need for a system arises that would be able to handle real-time data processing and analytics. This type of processing provides results as and when it happens. The most common method is to take the data directly from its source, which may also be referred to as a stream, and draw conclusions without actually transferring or downloading it. Another major technique in real-time processing is Data virtualization techniques where meaningful information is pulled for the needs of data processing

- 6) **Distributed Data Processing:** - Distributed data processing (DDP) is a technique for breaking down large datasets and storing them across multiple computers or servers. In this type of processing the task is shared by several resources/machines and is executed in parallel rather than being run synchronously and arranged in a queue. Because the data is processed in a shorter period, it is more cost-effective for businesses and allows them to move more quickly. Also, the fault tolerance of a distributed data processing system is extremely high.
- 7) **Multi-Processing:** - Multiprocessing is a type of data processing in which two or more processors work on the same dataset at the same time. In this multiple processors are housed within the same system. Data is broken down into frames, and each frame is processed by two or more CPUs in a single computer system, all working parallel.
- 8) **Time-Sharing Processing:** - The central processing unit (CPU) of a large-scale digital computer interacts with multiple users with different programs almost simultaneously in this type of processing. It is possible to solve several discrete problems during the input/output process because the CPU is significantly faster than most peripheral equipment (e.g., printers and video display terminals). The CPU addresses each user's problem sequentially, but remote terminals have the impression that access to and retrieval from the time-sharing system is instantaneous because the solutions are immediately available as soon as the problem is fully entered.

5.3 Acoustical data structure

DEF: Acoustics is defined by as "(a) Science of sound, including its production, transmission, and effects, including biological and psychological effects. (b) Those qualities of a room that, together, determine its character with respect to auditory effects."

The study of acoustics revolves around the generation, propagation and reception of mechanical waves and vibrations

Refer yourself back to Data Structure & Algorithm

5.4 File generation and management

FILE GENERATIONS: - are backup copies of a file that are created at specific times. Cataloging enables you to automatically keep track of files you might want to save for the following reasons: To keep one or two backup copies of a file.

FILE MANAGEMENT: - refers to the process of organizing and storing information in an easily retrievable manner. It involves creating an organized structure for file storage and using computer storage devices, such as drives, to hold information. By implementing an effective file management system, businesses can reduce clutter, improve organization, and enhance productivity.

The purpose of file management is to organize, store, and access electronic files efficiently and effectively. Good file management practices enable users to easily locate and retrieve the desired files, ensure data security, and optimize the use of storage space

5.5 File Sorting & Merging.

SORT statement: - Accepts input (from a file or an internal procedure) that is not in sequence, and produces output (to a file or an internal procedure) in a requested sequence. You can add, delete, or change records before or after they are sorted.

MERGE statement: - Compares records from two or more sequenced files and combines them in order. You can add, delete, or change records after they are merged.)

A program can contain any number of sort and merge operations. They can be the same operation performed many times or different operations. However, one operation must finish before another begins.

With Enterprise COBOL, your IBM® licensed program for sorting and merging must be DFSORT or an equivalent. Where DFSORT is mentioned, you can use any equivalent sort or merge product.

COBOL programs that contain SORT or MERGE statements can reside above or below the 16 MB line.

The steps you take to sort or merge are generally as follows:

- ❖ Describe the sort or merge file to be used for sorting or merging.
- ❖ Describe the input to be sorted or merged. If you want to process the records before you sort them, code an input procedure.
- ❖ Describe the output from sorting or merging. If you want to process the records after you sort or merge them, code an output procedure.
- ❖ Request the sort or merge.
- ❖ Determine whether the sort or merge operation was successful.

Restrictions:

- ❖ You cannot run a COBOL program that contains SORT or MERGE statements under z/OS UNIX. This restriction includes BPXBATCH.
- ❖ You cannot use SORT or MERGE statements in programs compiled with the THREAD option. This includes programs that use object-oriented syntax and multithreaded applications, both of which require the THREAD option. In addition, the COBOL program that uses SORT or MERGE statements cannot call directly or indirectly other applications that require z/OS Unix System Services or applications that use multithreading. For example, the JVM uses both of them.